CXR-LT 2026: Multi-Label Long-Tailed Classification on Chest X-Rays

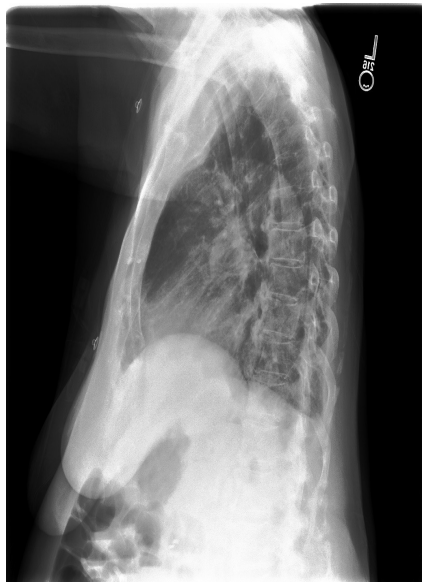
MIMIC-CXR-JPG-2.0.0



Postero-anterior : 64470



Anterior-postero : 100024

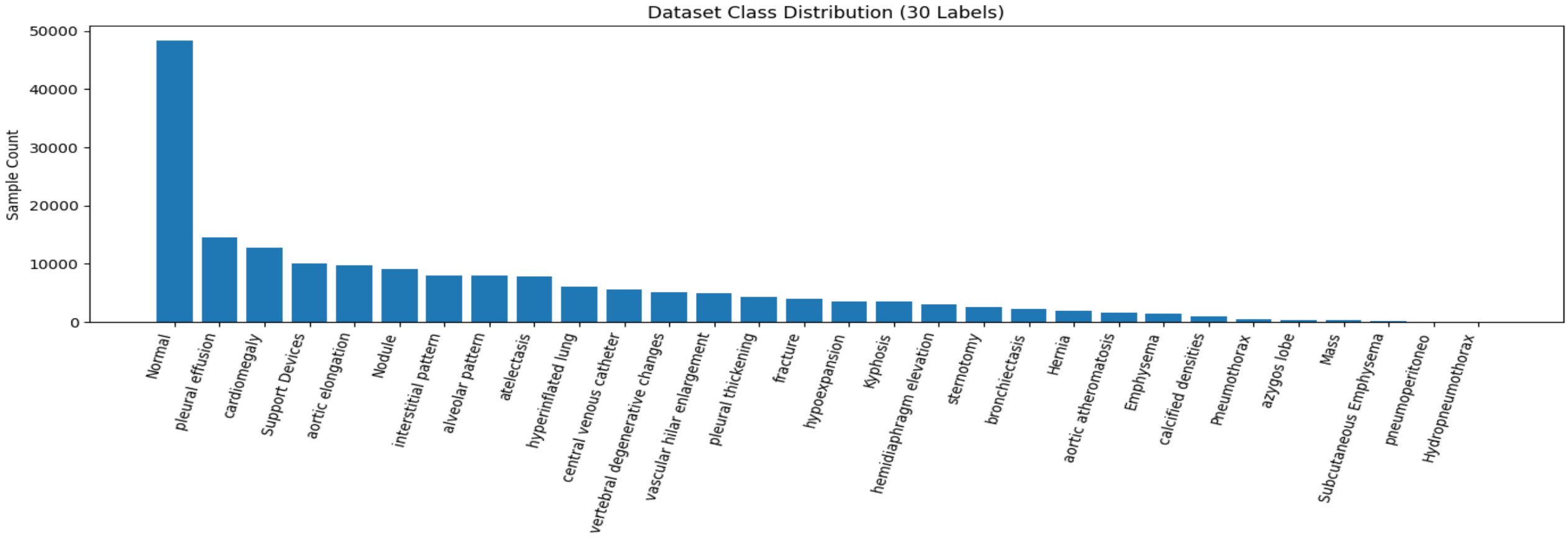


Left-lateral : 24080



Lateral : 57546

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Challenges

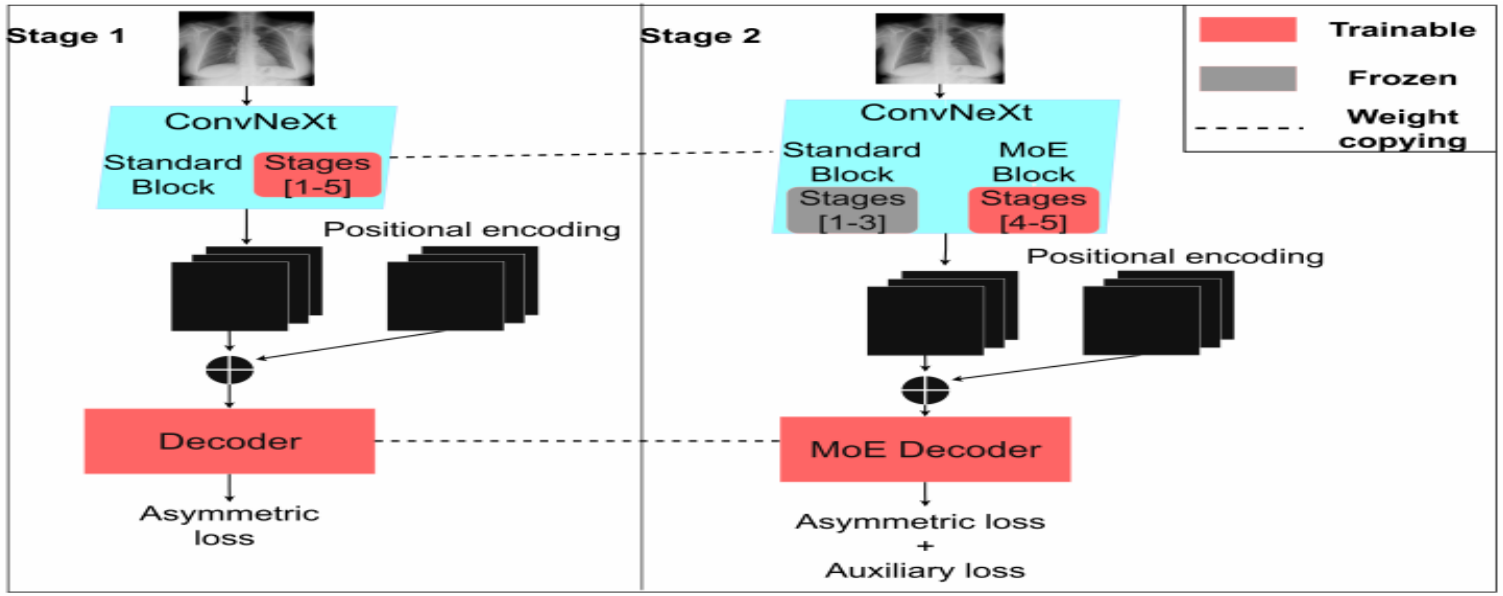
1. Label co-occurrence
   ⇒ The tail disease often co-exist with head diseases which makes the learning the individualized signatures of the tail cases challenging

2. Imbalanced data distribution

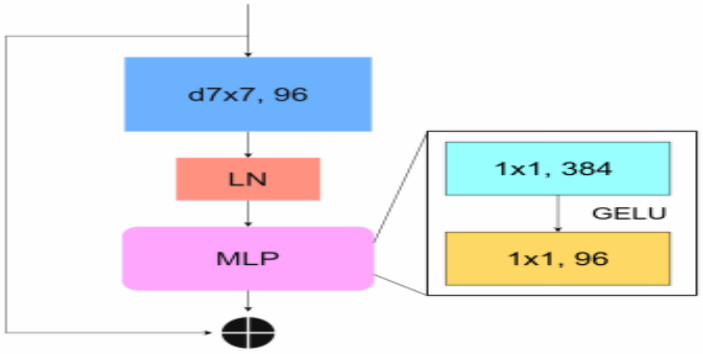
   ⇒ Inter-class: caused by long-tail data distribution

   ⇒ Intra-class: negative classes dominate the multi-label classification, model is penalized heavily for false positives

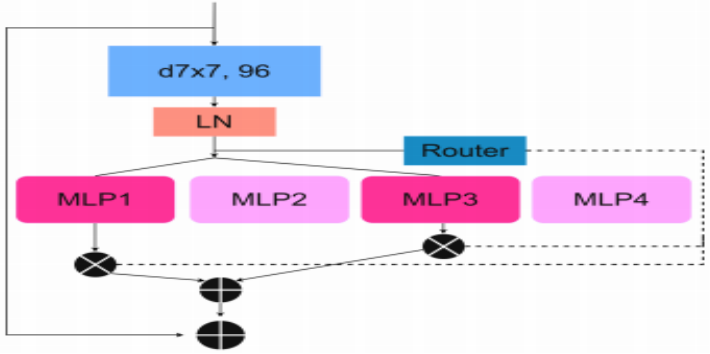
Our Old Proposed Framework



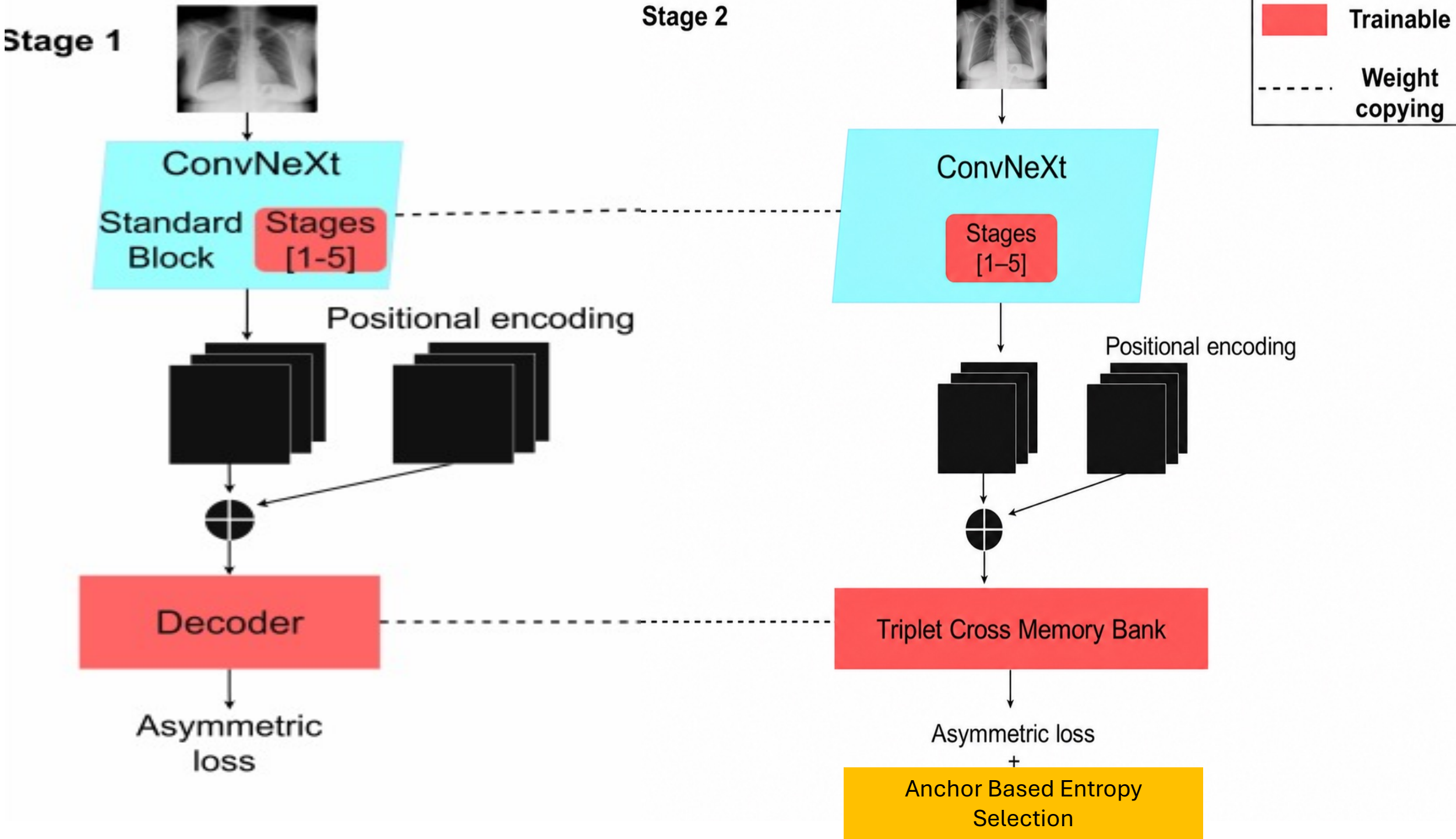
(a) Two-stages of the framework



(b) Standard Block



(c) MoE Block


Stage 1
Stage 2
Trainable
Weight copying
ConvNeXt
Standard Block
Stages [1-5]
ConvNeXt
Stages [1–5]
Positional encoding
Positional encoding
Decoder
Triplet Cross Memory Bank
Asymmetric loss
Asymmetric loss
+
Anchor Based Entropy Selection

Triplet Cross Memory Bank (Hardest Positive & Hardest Negative)

1) Current Mini-Batch (size = B)


x_1
[1, 0, 1, 0, 1, ...]


x_2
[0, 1, 0, 1, 0, ...]

⋮


x_B
[0, 0, 1, 0, 1, ...]

Encoder
$f(\cdot)$

Batch Embeddings

$E_{batch} \in \mathbb{R}^{B \times D}$

e_1 (D-dim)
e_2 (D-dim)
⋮
e_B (D-dim)

2) Cross Memory Bank (Global)

$M \in \mathbb{R}^{N \times D}$ (N >> B)

m_1 (label: [1, 0, 1, 0, ...])
m_2 (label: [1, 0, 1, 0, ...])
m_3 (label: [0, 1, 0, 1, ...])
⋮
m_N (label: [0, 0, 1, 0, ...])

Updated with Momentum
(enqueee / dequeue)

3) Distance Matrix

$dist \in \mathbb{R}^{B \times N}$

	m_1	m_2	m_3	...	m_N
e_1	0.21	0.13	0.84	...	0.65
e_2	0.72	0.61	0.18	...	0.90
⋮	⋮	⋮	⋮	...	⋮
e_B	0.33	0.71	0.26	...	0.77

Lower = More similar
(Cosine Distance)

4) For each anchor e_i (from batch)

Positive indices (same class)

pos_idx = [1, 2, 5, 8, ...]

dist[i][pos_idx]

0.21 | 0.13 | 0.19 | 0.28 | ...

argmin → index: 2

Negative indices (diff class)

neg_idx = [3, 4, 6, 7, ...]

dist[i][neg_idx]

0.84 | 0.73 | 0.91 | 0.55 | ...

argmax → index: 6

Hardest Positive
(min distance)
m_2

Hardest Negative
(max distance)
m_6

How we select:

```
pos_probs = dist[i][pos_idx]
neg_probs = dist[i][neg_idx]

hardest_pos = mem_emb[pos_idx[torch.argmin(pos_probs)]]
hardest_neg = mem_emb[neg_idx[torch.argmax(neg_probs)]]
```

Anchor
(from batch)
e_i

Hardest Positive
(closest same-class)
m_{pos} (= m_2)

Hardest Negative
(furthest diff-class)
m_{neg} (= m_6)

Small distance
$d(a, p)$

Large distance
$d(a, n)$

Overall Training Flow



5) Triplet Loss

$$L = \max(0,\ d(a, p) - d(a, n) + \text{margin})$$

$d(a, p)$: distance to hardest positive (small)
$d(a, n)$: distance to hardest negative (large)

Key Points

- ✓ Cross Memory Bank → large pool
- ✓ Hardest Positive = min distance same class
- ✓ Hardest Negative = max distance different class
- ✓ Makes triplet learning stronger

Triplet Cross Memory Bank + Anchor-Based-Entropy Selection

Why should integrate Anchor-Based-Entropy Selection?

How good the created Triplet? How good the anchor has selected?

Shannon Entropy Equation:

$$x_j^* = \arg\max \left(- \sum_i P_\theta(y_i|x) \log P_\theta(y_i|x) \right)$$

The anchor has selected by highest Shannon Entropy

Experimental Setup

Train and Val sets splitting	Sample size
Training set	142000
Internal Val set	17000

Performance on internal development set
Batch size = 8

Image Size = 384

Stage	mAP	mF1	mAUC	mECE
1	0.385	0.395	0.875	0.134
2	0.392	0.373	0.881	N/A

Image Size = 512

Stage	mAP	mF1	mAUC	mECE
1	0.385	0.395	0.875	0.134
2	0.405	0.395	0.884	N/A

Performance on internal development set
Batch size = 32

Image Size = 384

Stage	mAP	mF1	mAUC	mECE
1	0.385	0.395	0.875	0.134
2	0.405	0.421	0.884	N/A

Image Size = 512

Stage	mAP	mF1	mAUC	mECE
1	0.385	0.395	0.875	0.134
2	0.419	0.410	0.887	N/A





Thank you for your attention